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METHODS AND APPARATUS FOR SUPERTORREFACTION OF BIOMASS

Abstract

Under current practices, agricultural or landscaping waste left in a field or forestry waste left in a forest will decay and release greenhouse gases. In addition, forestry waste also poses a high risk for fires. Accordingly, mechanisms are provided to allow efficient conversion under anaerobic conditions of biomass, such as agricultural or forestry waste, into biocarbon product, such as biochar, biocoal, inert carbon and/or activated carbon, using molten salts as a more efficient heat transfer medium than conventional heat. Specifically, biomass is converted into biocarbon product during a supertorrefaction process during which molten salts are pumped under anaerobic conditions from a molten salt reservoir to a batch process cooker that holds the biomass. The salts are washed from the resulting biocarbon product as needed.

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Background/Summary

RELATED ART

[0001] During supertorrefaction, several physical and chemical changes occur in the biomass including the removal of moisture, volatiles, and hemicellulose, resulting in a material with higher energy density and increased carbon content. Depending on the operating condition, the output of supertorrefaction can be a range of biocarbon products from those with a highly energy-dense solid biofuel, often referred to as biocoal, to a less energy containing product for agriculture such as biochar or an inert carbon product suitable for carbon sequestration. This material has improved handling and storage properties compared to raw biomass and can be used in multiple applications like power generation and industrial heating, agriculture or carbon sequestration. Thus, the process is tunable by selecting operating conditions to produce a range of products.

[0002] Under some operating conditions, the supertorrefaction process can help reduce carbon emissions when biofuel produced by the process is used in place of fossil fuels. It is considered a carbon-neutral or even carbon-negative technology because the carbon dioxide (CO₂) released during combustion of the biofuel is offset by the carbon captured from the biomass through production of the biofuel. Under other operating conditions, the supertorrefaction process can produce carbon rich products suitable for application in agriculture or carbon sequestration either above or below ground.

[0003] Supertorrefaction has gained attention as a sustainable approach to biomass utilization and a means of producing a renewable and cleaner-burning solid fuel. It offers advantages in terms of energy density, transportability, and reduced greenhouse gas emissions when compared to untreated biomass. This application describes further improvements to the methods and apparatus that enable the use of the technology to produce products efficiently on a large scale by reducing the amount of energy and resources required to treat and convert the biomass into biocarbon product such as biocoal as biofuels or as biochar for agricultural use or for inert carbon for sequestration above or below ground.

Overview

[0004] Under current practices, agricultural or landscaping waste left in a field or forestry waste left in a forest will decay and release greenhouse gases. In addition, forestry waste also poses a high risk for fires. Because greenhouse emissions and fire risks negatively impact the environment, including contributing to negative impacts from climate change, effectively treating and processing biomass such as agricultural and forestry waste can significantly reduce these negative impacts. Accordingly, mechanisms are provided to allow efficient conversion of biomass, such as agricultural or forestry waste, into biocarbon products, such as biochar, biocoal, inert carbon, and/or activated carbon under anaerobic conditions using molten salts as a heat transfer medium. Specifically, biomass is converted into biocarbon products through a supertorrefaction process wherein molten salt is pumped from a molten salt reservoir to a batch process cooker that holds the biomass. In particular, molten salt is distributed through a plenum lid located on top of the cooker to flow vertically through the biomass before being collected in a molten salt reservoir located below the cooker.

[0005] The operating conditions of the batch process cooker and plenum lid can be varied to produce biocarbon products with the desired chemical and physical properties suitable for specific applications. The composition of the molten salts can also be varied to produce the biocarbon product suitable for specific applications. During treatment of the biomass with the molten salt, the

biomass is converted into biocarbon products such as biochar, biocoal, activated carbon, or the like. The processed biocarbon products are then washed in a sluice washer to remove excess salt deposited during the supertorrefaction process. The process can be enclosed, process intermediates are captured, energy and water are conserved. There is minimal emission. The enclosed system enables conservation of resources and management of processor steps by automation which respond to and sensors for process parameters to feedback information for the automation. Under some settings, the process can also be left open and the flexibility in the apparatus settings allows alternative settings to be chosen.

[0006] The resulting biocarbon products, such as washed biochar, biocoal, inert carbon, and/or activated carbon, can be used as a more carbon neutral, cleaner-burning source of energy or stored in a carbon capture environment that provides a carbon negative result. In some examples, the biocarbon products can be sequestered and stored in the ground or other location so that these products do not produce greenhouse gases or other negative environmental effects. In other examples, the biocarbon products can be used as a cleaner burning fuel with an overall carbon neutral or carbon negative impact on the environment. In the following disclosure, various illustrative, non-exclusive examples of inventive features relating to improved mechanisms and processes are described with reference to the figures below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The disclosure may best be understood by reference to the following description taken in conjunction with the accompanying drawings, which illustrate particular example embodiments.

[0008] FIG. 1 is a diagrammatic representation of an example of a supertorrefaction system.

[0009] FIG. 2 is a diagrammatic representation of an example of a cylindrical cooking system.

[0010] FIG. 3 is a diagrammatic representation of an example of a plenum lid.

[0011] FIG. 4 is a diagrammatic representation of an example of a biomass basket.

[0012] FIG. 5 is a diagrammatic representation of an example of a sluice washer.

[0013] FIG. 6 is a flow process diagram of an example of a supertorrefaction process.

[0014] FIG. 7 is a diagrammatic representation of an example of a control system used to automate the supertorrefaction process.

DETAILED DESCRIPTION

[0015] In the following description, specific details are set forth to provide illustrative examples of the systems and techniques described herein. The presented concepts may be practiced without some, or all, of these specific details. In other instances, well known process operations have not been described in detail to avoid unnecessarily obscuring the described concepts. While some concepts will be described with the specific examples, it will be understood that these examples are not intended to be limiting.

[0016] Supertorrefaction is a thermal treatment process that involves heating biomass materials, such as wood, agricultural residues, or other organic matter, in a low-oxygen environment submerged in molten salts to transform raw biomass into a range of biocarbon products, including those with more energy-density to those that are carbon intense stable inert carbon products. One key benefit of this process is that biomass is heated in an environment with limited oxygen to prevent combustion and reduce the release of greenhouse gases. Supertorrefaction involves heating biomass at temperatures significantly higher than those used in traditional torrefaction. Compared to torrefaction, supertorrefaction involves a shorter residence time of a few minutes within a reactor, instead of conventional torrefaction which requires times from at least 30 minutes but generally to more than a few hours.

[0017] Conventional torrefaction products that are minimally treated at 30 minutes have significant

levels of remaining greenhouse generating potential to render them basically untreated biomass. The shorter processing time by supertorrefaction allows for faster and greater devolatilization and thermal decomposition of the biomass to remove the potential for formation of greenhouse gases. Supertorrefaction has operating parameters that can be defined, controlled and monitored for performance. The range in operating conditions in supertorrefaction also means the process is tunable, where the preselected conditions such as salt composition, temperature, contact time, and other biomass treatment variables can be selected to shape the composition of the final product. The current application extends the range of operating conditions to add to those described in earlier filings. The result is also the ability to produce more carbon intense product than was possible with the earlier operating conditions. Various embodiments of the present invention also relate to monitoring, controlling, and automating supertorrefaction processes.

[0018] According to various embodiments, a supertorrefaction system includes a cooker having a plenum lid that distributes the molten salts onto the biomass in the cooker in a manner that converts the biomass into biocarbon product, such as biochar, biocoal, inert carbon and activated carbon. Any individual or mixture of a variety of carbon rich materials derived from biomass, agricultural waste, or organic waste used for carbon sequestration or as a substitute for fossil-derived carbon-based materials is referred to herein as biocarbon product. Some examples of biocarbon product include biochar, biocoal, inert carbon, activated carbon, biomass pellets, and bio-based chemicals.

[0019] The resulting biochar or other biocarbon products can then be washed to various degrees using a cleaning sluice and either sequestered, used as soil amendments or reused as a renewable biofuel in its more energy intense and clean burning form. The composition of the salts can be any eutectic salt mixture and the operating conditions of the batch process cooker and plenum lid can be varied in a tunable process to different temperatures, contact times and salt combinations to produce biocarbon products with the desired chemical and physical properties suitable for specific applications. The process intermediates are captured and condensed, energy and water can be recycled within the process. The process can be fitted with sensors and automated through the use of computer controls.

[0020] With reference to FIG. 1, shown is a diagrammatic representation of an example of a supertorrefaction system **101**. The supertorrefaction system **101** efficiently converts biomass into biocarbon products while addressing environmental concerns and optimizing resource utilization. Supertorrefaction system **101** includes a batch process cooker **103**, such as a cylindrical cooker, a molten salt reservoir **105**, a molten salt pumping system **107**, a sluice washer **109** or other washer for salty biocarbon product, and a VOC (Volatile Organic Compounds) management system **121**.

[0021] According to various embodiments, a batch process cooker **103** is a core component of the system, where biomass **151** is thermally treated at high temperatures and low oxygen levels to produce biocarbon products. In the present example, biomass **151** undergoes a supertorrefaction process in batch process cooker **103** until it is converted into hot salty biocarbon product **111**. Specifically, once the biomass **151** is placed in batch process cooker **103**, hot molten salt is distributed onto biomass **151** within batch process cooker **103**, such that the molten salt flows through biomass **151** and provides the heat necessary to convert the biomass into biocarbon product **111**. In the present example, hot molten salt is pumped from a reservoir **105** into batch process cooker **103** using pump **107**. The molten salt is heated in reservoir **105**, which includes an electrical resistive heating element **110** that keeps the molten salt at a desired temperature necessary for the supertorrefaction process. Once the molten salt flows through and out of the batch process cooker **103**, it is recirculated into the reservoir of molten salt **105** to be reheated and recirculated again into the batch process cooker **103**.

[0022] In particular embodiments, the hot salty biocarbon product **111**, which has been produced by the supertorrefaction process in the batch process cooker **103**, is removed using a robotic arm that either moves a basket holding the salty biocarbon product **111** from the batch process cooker **103** to a sluice washer **109** or moves the biocarbon product **111** from the batch process cooker into

a basket in the sluice washer **109**, such as by tipping the batch process cooker or a vessel within the batch process cooker so that the biocarbon product **111** falls into the basket in the sluice washer. [0023] In the present example, the sluice washer **109** is used to wash the hot salty biocarbon product **111** and remove salts and other contaminants from the biocarbon product **111**. As shown, water **137** passes into sluice washer **109** and flows out of the sluice washer **109** as quench water **115**, which is saturated with salt from washing the hot salty biocarbon product **111**. The quench water **115** can be directed back into the reservoir of molten salt **105** to replenish salt levels and maintain the system's functionality. In the present embodiment, the salt saturated quench water can be dripped **117** into the reservoir of molten salt **105**. This drip **117** can be used to direct the desired salt into reservoir **105** without including excess water or contaminants. In addition, hot steam **119** generated in the system **101** can be routed back to a condenser **127** or used for other heating applications within the system. The washed biocarbon product **100** can be collected for use outside of system **101**. For instance, the washed biocarbon product **100** can be sold as carbon offsets to companies or other entities that produce excess greenhouse gas emissions in order to reduce the net greenhouse gases emitted.

[0024] In particular embodiments, VOCs are generated as byproducts of the supertorrefaction process in the cylindrical cooker **103**. To remove the volatile organic compounds (VOCs) **131** generated during the supertorrefaction process, an exhaust pipe connects the cooker **103** to a VOC management unit **121** that collects and processes the VOCs **131**. This VOC management system may include one or more spargers **123**, VOC oxidizers **125**, and condensers **127**.

[0025] In the present example, the VOCs **131** are extracted from cooker **103** and routed for further processing. According to various embodiments, spargers **123** or exhaust systems are used to capture the VOCs **131** and control the airflow within the supertorrefaction chamber. Specifically, the spargers **123** direct VOC-laden gases to the next stages of treatment.

[0026] In particular embodiments, the VOC oxidizer **125**, often in the form of a scrubber, is responsible for converting the VOCs **131** into less harmful compounds by oxidizing them. This helps prevent the formation of tar precursors and reduces emissions. According to various embodiments, after passing through the oxidizer, the VOCs **131** are converted into water vapor (H.sub.2O) and carbon dioxide (CO.sub.2) **129**, which are less harmful and easier to manage.

[0027] In the present example, a condenser **127** cools down and condenses the hot gases, using cooling water **133**. The condensed products include condensed water **135**, now in liquid form, and CO.sub.2 that is released into the atmosphere. The condensed water **135** can be recirculated within the system depending on the desired implementation. For instance, the condensed water **135** can be used for one or more purposes, such as washing the salty biocarbon product, cooling the equipment, or other thermal management tasks. As shown in the present example, the condensed water **135** can be directed to be used as water for washing biocarbon product **137**.

[0028] As described, the integrated supertorrefaction system **101** not only efficiently converts biomass **151** into biocarbon product **100** but also manages VOC emissions, addresses environmental concerns, and maximizes resource utilization by recycling and repurposing water and heat energy. It demonstrates a comprehensive approach to sustainable biomass processing while minimizing environmental impact. In particular, the net effect of processing biomass **151** into biocarbon product **100** with system **101** is to significantly reduce the amount of greenhouse gases that would have been produced if the biomass **151** were instead allowed to decay on its own.

[0029] By varying the operating temperature, the composition of the salts, the duration of supertorrefaction, and the variety of the incoming biomass, a range of biocarbon products may be produced, rendering the process tunable to the desired application. It should be noted that a variety of salts and salt combinations may be used in system **101**. For instance, any combination of eutectic salts is within the scope of the invention.

[0030] With reference to FIG. 2, shown is a diagrammatic representation of one example of a cylindrical cooking system **200** that can be used in a supertorrefaction process. In the present

embodiment, cylindrical cooker **201** can be used as a batch process cooker to transform biomass into biocarbon product via a supertorrefying batch process. In some examples, cylindrical cooker **201** can include a porous basket and may be situated as a cylindrical stationary cooker. According to various embodiments, the cylindrical cooker **201** itself is a robust, cylindrical vessel made of heat-resistant materials, such as stainless steel or specialized alloys. It is designed to withstand high temperatures and corrosive environments.

[0031] According to various embodiments, the cooking system **200** includes a hoist system that facilitates inserting a portable porous basket filled with a batch of biomass into cylindrical cooker **201**. The holding basket remains within the cylindrical cooker **201** during the supertorrefaction process. In some implementations, the hoist system can also be used to remove the holding basket from the cylindrical cooker after the biomass has been converted to biocarbon products.

[0032] In particular embodiments, the bottom of the cylindrical cooker **201** features a mesh drain **203**, perforated plate, or the like, which is designed to allow molten salt to pass through the cylindrical cooker **201** while allowing the biomass and/or biocarbon product to remain within the cylindrical cooker **201** throughout the supertorrefaction process. According to various embodiments, the mesh drain **203** is used to facilitate the controlled flow of molten salt out of the cylindrical cooker **201** and into a molten salt reservoir tank **205** that is located below the cylindrical cooker **201**. Salts are collected, heated, and maintained at a specified temperature in a molten state in the molten salt reservoir tank **205**. The molten salt reservoir tank **205** may be constructed to handle the high temperatures and corrosive nature of molten salt. According to various embodiments, the molten salt reservoir tank **205** includes one or more electrical resistive heating elements **215** used to melt the salt to a fluid state and to maintain it at a constant predetermined temperature, for example 200-700 degrees Celsius. In particular embodiments, this molten salt is pumped vertically through a supply pipe connecting a molten salt reservoir **205** to the cylindrical cooker **201** or other cooking structure through a plenum lid **209** that is located above the cylindrical cooker **201**. The molten salt flows from the plenum lid onto the biomass within the cylindrical cooker **201**. In particular, molten salt is evenly distributed on top of the biomass, such that it flows through the basket of biomass and transfers its heat to the biomass in a manner that converts it into biocarbon product through supertorrefaction. A filtration mesh apparatus, such as mesh drain **203**, located at the bottom of the supertorrefaction unit keeps the biomass and resultant biocarbon product within cylindrical cooker **201** while allowing the molten salt liquid to pass through and drain into the underlying molten salt reservoir **205** to be re-heated and recycled through the supertorrefaction system.

[0033] According to various embodiments, a pump system **207** is connected to the molten salt reservoir and is responsible for circulating the molten salt into the cylindrical cooker **201**. In the present example, the pump system **207** includes a pump and pipes connecting the molten salt reservoir **205** to the plenum lid **209** above cooker **201**. The pump drives the molten salt heat transfer liquid through a pipe that connects the reservoir **205** to plenum lid **209** of the cooker. In particular embodiments, the plenum lid **209** is designed with an array of holes at its base through which the molten salt heat transfer liquid flows, such that the molten salt is uniformly distributed over the biomass located in cooker **201**. The plenum lid **209** may include features such as distribution channels, nozzles, a network of pipes, or the like, that help to evenly distribute the molten salt on top of the biomass within the cylindrical cooker **201**. In the present example, molten salt flows vertically from the plenum lid **209** through a basket within cooker **201** to directly contact the biomass within the basket. As the hot molten salt passes through the biomass and out of the cylindrical cooker **201** through drain **203**, heat is transferred to the biomass at a temperature that allows the biomass to be converted into biocarbon product through a supertorrefaction process.

[0034] In particular embodiments, pump system **207** includes a bypass system **211**. This system **211** allows the pumped molten salt to be redirected back into the reservoir **205** when the cylindrical cooker **201** is open, such as when biomass is loaded or when biocarbon product is being removed

from cylindrical cooker **201**. For instance, during periods when a basket of loaded biomass is inserted into the cylindrical cooker **201** and/or a basket of biocarbon product is removed from the cooker **201** for washing and cooling, the flow of hot molten salt into the cooker **201** needs to be paused. In particular, the molten salt flow to the cooker is bypassed and redirected by a shunt valve **213** from the reservoir tank **205** through bypass **211** and back into the reservoir tank **205**. In the present example, the shunt valve **213** diverts the molten salt liquid flow to bypass the cooker and routes the liquid directly back into the reservoir **205** during two time periods. Specifically, recirculation of fluid within the bypass system **211** takes place when (i) biomass is loaded into the receiving cooker, such as when biomass is loaded into a basket within the cooker or when a basket holding biomass is loaded into the cooker, and (ii) when a basket of salty biocarbon product is removed from the cooker and unloaded into a porous washing basket for cleaning and cooling. In the present example, recirculation of the molten salt within the bypass system **211** helps the molten salt to maintain a consistent temperature and prevents unnecessary heat loss during loading and unloading of biomass and biocarbon product batches.

[0035] According to various embodiments, the cooking system **200** can also include a VOC port **213** that allows VOCs produced during the supertorrefaction process to flow out of the system **200** for further treatment. For instance, the VOC port **213** may be routed to a VOC management system **121**, as described above with regard to FIG. **1**. There, the VOCs are processed into less harmful substances before being released into the atmosphere. Alternatively, the VOCs may also contain substances of value that can be captured and used in other applications.

[0036] In the present example, the supertorrefaction process involves pumping molten salt from reservoir **205** to plenum lid **209** that evenly distributes the molten salt onto the top of the biomass located inside a container within the cylindrical cooker **201**. In some examples, the container may be a cylindrical basket, such as a mesh basket, a cylindrical vessel with a mesh bottom, etc. The molten salt flows through the biomass, effectively heating and converting it into biocarbon product. The entire process is controlled and monitored by a computer-based control system. This system regulates the temperature, flow rate, and timing of the circulation of molten salt, ensuring optimal supertorrefaction conditions. The system also controls the recirculation of molten salt in the bypass system, which increases the overall energy efficiency and safety of the biomass supertorrefaction process. The system can also be used to create an integrated system linked to the washing system.

[0037] Although the present example describes using a cylindrical cooker **201**, it should be noted that other shapes of cookers may be used depending on the desired implementation. Accordingly, an elliptical, rectangular, square, barrel-shaped, spherical, or other shaped cooker may be used in some applications without departing from the scope of the present disclosure. Furthermore, although the present example describes a stationary cooker, moveable cookers, such as moveable spherical cookers, may be used in some implementations.

[0038] With reference to FIG. **3**, shown is a diagrammatic representation of an example of a plenum lid. As described above with regard to FIGS. **1** and **2**, a plenum lid **209** can be used as a vessel to distribute molten salt onto biomass as part of a supertorrefaction process to convert the biomass into biocarbon products such as biochar, biocoal, inert carbon, and/or activated carbon. According to various embodiments, the plenum lid **209** is designed as a flat, circular, rectangular, or other-shaped structure that covers the top opening of a cylindrical cooker **201**. The lid **209** is typically made of heat-resistant materials, such as stainless steel, capable of withstanding high temperatures and exposure to molten salt. According to various embodiments, a wide range in temperatures, such as 200° C. to 700° C. can be used. In some embodiments, the plenum lid **209** includes an intricate network of distribution channels, pipes, nozzles, or the like, that allows the molten salt to be evenly spread and/or sprayed over the surface of a batch of biomass **151** located within the cylindrical cooker **201**.

[0039] According to various embodiments, the distribution channels or nozzles **313** allow the molten salt **321** to be distributed uniformly across the entire top surface of the biomass **151**. This

even distribution is important for achieving consistent heating and conversion of the biomass into biocarbon product. In particular embodiments, some plenum lids may have adjustable mechanisms or flow control valves that allow operators to fine-tune the distribution of molten salt based on specific process requirements.

[0040] In the present example, the plenum lid **209** is connected to a molten salt supply system. Molten salt is pumped into the lid **209** from a reservoir of molten salt and is directed into the distribution channels or nozzles **313**. In particular embodiments, the plenum lid **209** relies on gravity to facilitate the flow of molten salt from the distribution channels onto the biomass below.

[0041] According to various embodiments, the plenum lid **209** helps maintain a consistent temperature within the cylindrical cooker, as the molten salt not only heats the biomass but also transfers heat to structures within the cylindrical cooker throughout the process. Maintaining a consistent temperature within the cooker can improve the quality and efficiency of the supertorrefaction process. For instance, by maintaining a consistent temperature within the cylindrical cooker, the biomass can be uniformly processed at the same temperature throughout the batch.

[0042] In the present embodiment, the plenum lid **209** is integrated into the overall control system of the supertorrefaction equipment. It receives commands and instructions regarding the timing, flow rate, and duration of molten salt distribution. Temperature and pressure sensors **311** may be installed in the lid to provide real-time data to the control system, allowing for precise monitoring and adjustment of the supertorrefaction process. Safety interlocks may be incorporated into the lid's design to ensure that the distribution of molten salt can be immediately stopped in case of emergencies or system malfunctions. In some implementations, the plenum lid **209** may also include heat insulation or heat-resistant coatings to protect operators and maintain the integrity of the lid structure.

[0043] With reference to FIG. 4, one example of a biomass basket is shown as part of a blow-apart diagram. In this example, biomass basket **401** is shown relative to a cylindrical cooker **201** and a plenum lid **209**. As described above with regard to FIGS. 1-3, the cylindrical cooker **201** and plenum lid **209** can be used together to hold and process a batch of biomass **151** into biocarbon product during a supertorrefaction process.

[0044] According to various embodiments, the biomass basket **401** is constructed from heat-resistant materials such as stainless steel capable of withstanding the extreme temperatures. The biomass basket **401** is built to withstand the extreme temperatures of a molten salt supertorrefaction process without warping, corroding, or deteriorating. In particular embodiments, the biomass basket **401** is cylindrical with solid sides and a strainer bottom **403** designed to allow molten salt **321** to flow from a plenum lid **209** through the biomass **151** and out of the biomass basket **401** through a strainer bottom **403**.

[0045] In the present example, the basket **401** is cylindrical in shape, resembling a tall, open-ended cylinder with a flat base. The size of the biomass basket **401** can be selected based on the specific application and the volume of biomass to be processed in conjunction with the molten salt. In particular embodiments, the sides of the basket are solid and made of a sturdy, perforation-free metal material. These solid sides are designed to contain the biomass material **151** and molten salt **321** during the supertorrefaction process.

[0046] According to various embodiments, the bottom of the basket **401** includes a strainer design **403** that allows molten salt **321** to flow through the bottom of the basket **401** into a molten salt reservoir. In particular embodiments, this strainer **403** includes evenly spaced holes, slots, mesh, porous material, or the like, specifically designed to facilitate the flow of molten salt through the basket **401** while retaining the biomass and/or processed biocarbon product inside the basket **401** throughout the supertorrefaction process.

[0047] In the present example, the top of basket **401** is connected to a plenum **209** having multiple openings or inlets through which molten salt is introduced into the basket **401**. In particular

embodiments, the inlets can be fitted with a mechanism for precise control of the molten salt flow rate. In other examples, the biomass basket may be nested within a stationary cylindrical cooker **201** and the plenum lid **209** may be positioned above, but not necessarily in contact with, the biomass basket. In yet other examples, the biomass basket and the cylindrical cooker may be combined into one structure such that the biomass basket serves as the cylindrical cooker **201**. In each of these examples, the cylindrical cooker may be stationary or moveable, depending on the desired implementation.

[0048] According to various embodiments, an attachment at the top of the biomass basket **401** is included to allow a robotic arm to remove the basket **401** from the cylindrical cooker **201** before and/or after processing a biomass **151** batch. Using a robotic arm to engage the biomass basket **401** allows for automation and safe handling of the basket **401** during loading of the raw biomass and removal of the processed biocarbon product or other solid materials. In particular embodiments, the top of the basket includes an interface designed to be compatible with the end effector or gripper of a robotic arm. This interface may include features such as mechanical connectors, fasteners, or a specific pattern for robotic arm attachment. The attachment is engineered to allow the robotic arm to securely attach to or grip and lift the basket with minimal risk of slippage or accidents. This often involves the inclusion of handles, hooks, or other suitable lifting mechanisms. The attachment may incorporate additional safety features like sensors or locking mechanisms to ensure that the basket is properly secured before being lifted. In particular embodiments, the attachment is reinforced to handle the weight of the loaded biomass basket **401**, which can be substantial, especially in industrial-scale processes.

[0049] In some embodiments, the attachment may have a control interface that allows the robotic arm to communicate with the attachment, ensuring proper engagement and disengagement. The attachment is designed to be compatible with the specific supertorrefaction chamber and robotic arm system in use, ensuring seamless integration and operation and facilitating the automation of the supertorrefaction process, improving efficiency, reducing the risk of accidents, and allowing for precise control over the timing and handling of the basket. According to various embodiments, remote operation can be implemented, which is particularly valuable in situations where the supertorrefaction chamber is located in a hazardous environment or where human access is limited or unsafe.

[0050] According to various embodiments, a transporting apparatus such as hoist system can be used to lift a basket **401** of processed biocarbon product from the supertorrefaction cooker **201** and/or tip the basket **401** to allow the biocarbon product to fall into a cylindrical washing chamber, where excess salt and other contaminants can be washed from the processed biocarbon product. It should be noted that in some embodiments, the basket **401** of biocarbon product is itself used as a washing chamber or is lifted and placed into a washing chamber or other washing location.

[0051] In particular embodiments, processed biocarbon product is washed of its salt content using a sluice washer, an example of which is shown and described with regard to FIG. 5. With reference to FIG. 5, a diagrammatic representation of one example of a sluice washer **501** is shown, which can be used to wash salty biocarbon product produced in a supertorrefaction process as described above with regard to various examples. In the present example, sluice washer **501** includes a cylindrical chamber **507** that rolls on a track **517** that can include rails or other guide mechanisms or structures, in an intermittent manner above and along the length of an open-ended sluice **515**. According to various embodiments, the sluice **515** is a channel through which water can flow. A pump **503** fixed at one end of the sluice **515** inputs fresh water, which flows horizontally through the sluice **515** and out through drain **505**. A portable pump **519** attached to a moveable washing chamber **507** is used to force water to flow vertically through the biocarbon product located within chamber **507**.

[0052] According to various embodiments, chamber **507** is a porous basket having solid metal sides, a wire mesh bottom, and a plenum lid. Wheels may be attached to the basket structure to aid

in its motion along track **517**, which extends above and along the length of sluice **515**. In some embodiments, the basket and plenum lid may be moved together from a supertorrefaction chamber to a wheeled platform that rolls along track **517**. The wheeled platform may have an open bottom that allows fluid to flow freely out of the basket and into the sluice **515**. In other embodiments, the wheels may be attached to the chamber before or after the supertorrefaction process. In yet other embodiments, the track may include wheels or rollers that allow the chamber to roll when placed on the track. For instance, when the chamber is placed on the track **517** it can interlock or otherwise engage with the track so that it can move in a guided manner along the length of the sluice **515**. In other examples, there may be walls on the sides of the rollers that guide the chamber when it glides across the rollers. In these and other embodiments, the rollers or wheels may not need to be attached or fixed to chamber **507**.

[0053] In the present example, freshwater, treated water, or condensation from other processes associated with supertorrefaction is supplied to the sluice washer **510** from a water source **513**. The water from water source **513** is continuously pumped by pump **503** from one end of the sluice **515** to the other end of the sluice **515** towards drain **505**. According to various embodiments, the sluice **515** can be a long, rectangular trough made of corrosion-resistant materials such as stainless steel. In some embodiments, the trough **515** is inclined at a gentle slope from a lower end near the drain **505** to a higher end near the pump **503** to facilitate the flow of water and salt along the trough **515** and out through the drain **505**. Depending on the desired implementation, the sluice **515** can also be shaped in different ways that may improve efficiency of water and salt flow along the length of the sluice. For instance, the sluice **515** may be shaped as a rectangular channel with a flat bottom in some applications, and may be shaped as a u-shaped channel with a curved bottom in other applications. Accordingly, the sluice **515** can be designed with the materials, dimensions (i.e., length, width, and depth) and shape (i.e., cross-sectional shape) to achieve the desired flow dynamics within the channel.

[0054] According to various embodiments, a basket **507** containing salt-laden biocarbon product is positioned above sluice **515** near the drain end **505** of the sluice washer **501**. The basket **507** gradually moves along track **517** towards the pump end **503** of the sluice washer **501**. The salt-laden biocarbon product receives a continuous injection of water from one or more nozzles or spray bars, which may be associated with one or more pumps **519**. For instance, water from sluice **515** can be pumped into a basket within chamber **507** using a portable pump **519**. In particular embodiments, the water is pumped into a plenum lid of chamber **507**. The plenum lid uniformly distributes the water through a basket of biocarbon product located within chamber **507**.

[0055] As water is pumped onto the salty biocarbon product within chamber **507**, it begins to dissolve the salts and carry away the salts present on the biocarbon product's surface and pores. The water becomes salty and flows out through the bottom of the basket in chamber **507** into sluice **515**. During this washing process, the local salinity of the water in the sluice **515** below chamber **507** increases as salt is washed from the biocarbon product and into the sluice **515**. Specifically, below each location where chamber **507** stops and goes through a wash cycle, the salinity of the water in sluice **515** increases. Because water in sluice **515** flows continuously from the pump **503** end (upstream) to the drain **505** end (downstream), the water upstream from chamber **507** in the sluice **515** remains clean and water downstream from chamber **507** becomes salty. As shown in FIG. 5, clean water pumped through chamber **507** with portable pump **519** is drawn up from sluice **515** on the upstream side of chamber **507**.

[0056] One advantage to moving the chamber along track **517** for successive washing cycles is that water drawn up from the sluice **515** for washing is less likely to be contaminated by salty water deposited into the sluice **515** by chamber **507** during previous washing cycles because the salty water is less likely to become increasingly concentrated near the chamber **507** at a given location in the sluice. If the chamber remains stationary in one location during successive wash cycles, salty water may accumulate at one location in sluice **515** and may take some time before the salinity at

this location dissipates as the water flows out towards drain **505**. Depending on the flow rates of the water in sluice **515**, salt may also accumulate on the bottom of the sluice and may slowly wash away with continuous flow of water through the sluice **515**. In the present embodiment, each washing cycle deposits more salt into sluice **515** and locations further downstream may have increasing concentrations of salt as it flows towards drain **505**. Consequently, by moving the chamber **507** upstream along track **517**, it is possible to continue pumping clean washing water into chamber **507** from sluice **515** throughout the washing process.

[0057] According to various embodiments, washing chamber **507**, which includes a basket to hold biocarbon product, can be mounted on a conveyor system or carriage with wheels, allowing it to be moved along the length of the sluice washer **501**. In particular embodiments, the washing chamber **507** begins at a position above the drain end of sluice **515** and moves along track **517** with the assistance of a drive mechanism, which includes drive bar **509** and ratcheted gear drive **511**. In the present example, the drive bar **509** is attached to the structure of the washing chamber **507** structure to promote its motion along rails **517** in an upstream direction. In some embodiments, the chamber **507** moves along track **517** intermittently in a sequential step-wise procedure that includes washing the biocarbon product while stopped at a first location on the track, then moving to second location that is upstream from the first location, and washing the biocarbon product while stopped at this second location. This procedure continues with the washing chamber **507** moving upstream along the length of the trough shaped structure **515** toward the pump end of the sluice **505**. However, in other embodiments, the chamber **507** can continuously move along track **517** during a continuous or intermittent washing process.

[0058] At one end of the sluice washer **501**, there is a drain system **505** to collect the salt solution deposited during the washing process and divert it to a separate recovery tank or treatment facility for further processing, reuse, or disposal. In some embodiments, the salt can eventually be reused upon its return to a molten salt reservoir, examples of which were previously described with regard to FIGS. **1** and **2**. By reusing the salt in the molten salt reservoir, the supertorrefaction system can operate more efficiently with less waste products.

[0059] In particular embodiments, an automation and control system allows operators to adjust and monitor various parameters, including water flow, conveyor speed, and the duration of the washing process. Additionally, the control system may allow operators to program the amount of washing desired. The system may be equipped with sensors that measure salinity at selected locations in the sluice **515** or at other locations in the sluice washer **501**, such as in chamber **507** or at the mesh drain of chamber **507**. Sensors can also be used to monitor other parameters within the sluice washer **501** depending on the desired implementation. In some embodiments, the sensors can be controlled by a computer program that is used to automate the wash process. The washing parameters implemented by the automation and control system may be adjusted to accommodate the type of incoming biocarbon product, the intended use of the biocarbon product, the type of salt and temperature used in the supertorrefaction process, and other parameters affecting how the biocarbon product was produced during the supertorrefaction process. The control system can be adjusted to the degree of washing desired as some end product uses can tolerate more residual salts than other end product uses.

[0060] According to various embodiments, safety measures are integrated into the design to ensure the safe operation of the equipment, including emergency stop mechanisms and guarding to protect personnel. In some embodiments, these safety measures can be implemented by the automation and control system. Additionally, these safety measures can include mechanical or physical features that can protect personnel independently of the operation of the automation and control system.

[0061] The present example of a sluice washer **501** demonstrates a system that can be used to implement a continuous and efficient process for washing salt off of biocarbon product produced by a supertorrefaction process. In particular, by pumping water into one end of the sluice, moving the biocarbon product basket along the length of the sluice, and collecting the saline solution at the

other end, the sluice washer **501** can effectively clean hot salty biocarbon product while also allowing reuse of water and salt in the supertorrefaction and washing processes. As described, the washing water can come from various sources, such as condensed water produced as a result of the supertorrefaction and/or VOC treatment processes. Furthermore, salt collected in the sluice **515** during the washing process can be recycled and reused as molten salt in the supertorrefaction process.

[0062] With reference to FIG. **6**, shown is a flow process diagram describing one example of a supertorrefaction process for efficiently converting biomass into biocarbon product. According to various embodiments, a system such as the one described above with regard to FIG. **1** is connected to a control unit that can be used to regulate and/or automate the supertorrefaction process. In the present example, a control unit, which is connected to various items within the system, receives input parameters such as the type of biomass, processing time, and temperature settings at **601**.

[0063] According to various embodiments, a robotic arm, guided by the control unit, loads a cylindrical supertorrefaction basket with biomass to be processed at **603**. The control unit monitors and regulates the molten salt heating process inside the supertorrefaction chamber, ensuring it adheres to the desired temperature and time parameters at **605**. Throughout the supertorrefaction process, the control unit continuously collects data from the sensors in components such as the basket, the chamber, the pump, the reservoir of molten salt, and the robotic arm.

[0064] In particular embodiments, the control system adjusts the temperature, pressure, and other parameters as needed to optimize the supertorrefaction process for efficient biocarbon product production at **607**. Once the supertorrefaction process is complete, the control unit instructs the robotic arm to retrieve the processed biocarbon product in the supertorrefaction basket. The robotic arm uses an attachment mechanism to securely grip and lift the basket.

[0065] The robotic arm, under the control of the central unit, moves the supertorrefaction basket to the sluice washer's loading area at **609**. In some implementations, the supertorrefaction basket is instead tipped so that the biocarbon product is poured out or otherwise transferred into a chamber in the sluice washer. In particular embodiments, the control unit initiates the washing process in the sluice washer at **611**. It controls water flow and the movement of the biocarbon product basket along a track from the drain end of the sluice to the pump end of the sluice. In some implementations, the control unit may also control aspects such as agitation of the sluice and the biocarbon product basket. According to various embodiments, sensors in the sluice washer and the robotic arm may provide real-time data about the washing process. The control unit ensures that the biocarbon product is thoroughly washed and substantially free of contaminants if the end use so requires. According to various embodiments, the degree of washing can be set to a level suitable for producing a particular biocarbon product. Biocarbon products for sequestration may require less washing while biocarbon products used for agriculture may require more washing. The amount of washing can be automated using sensors and feedback associated with the control system.

[0066] In particular embodiments, multiple lines may be included to allow for simultaneous processing at particular components. For example, multiple cookers and multiple VOCs may be included in a system having a single sluice washer. Alternatively, multiple sluice washers may be included in a system having a single cooker in order to increase throughput and efficiency.

[0067] According to various embodiments, the control system includes safety features such as emergency stop buttons and safety interlocks to ensure safe operations. It also logs data from all components, enabling operators to monitor the entire process and troubleshoot any issues. The system logs detailed data for each step of the process, which can be used for quality control, compliance reporting, and process optimization.

[0068] In the present example, once the basket holding the biocarbon product reaches the end of the sluice near the pump end, the washed biocarbon product can then be removed from the sluice washer at **613**. At this point, the process ends and the washed biocarbon product is the desired end product. According to various embodiments, the washed biocarbon product can then be sequestered

or used as fuel in a manner that is carbon neutral or carbon negative. As described in this example, the entire supertorrefaction process can be automated from the time the raw biomass enters the system to the time the washed biomass exits the system.

[0069] With reference to FIG. 7, shown is a diagrammatic representation of a computer control system. According to various embodiments, the control system **700** can be used to automate the supertorrefaction process described previously with regard to FIG. 6. In particular, a supertorrefaction system, such as the system described with regard to FIG. 1, can be managed and automated by a centralized computer-based control unit. This unit is responsible for coordinating all components of the automation process. For instance, it can control loading of raw biomass into the system, heating and distributing molten salt onto the biomass to convert the biomass to biocarbon product, moving the processed biocarbon product into a sluice washer, operating and controlling the sluice washer, and removing the washed biocarbon product from the washer. The control system **700** can be used to automate various parts of the process and monitor conditions of the system during processing.

[0070] According to various embodiments, control system **700** is a particular example of a computer system that can be used to implement particular examples of the present invention. For instance, the control system **700** can be used to implement a computing device, which can include a mobile device, computer, laptop, etc., to automate a supertorrefaction process according to various embodiments described above. In particular example embodiments, a system **700** suitable for implementing particular embodiments of the present invention includes a processor **701**, a memory **703**, an interface **711**, and a bus **715** (e.g., a PCI bus). The interface **711** may include separate input and output interfaces, or may be a unified interface supporting both operations. When acting under the control of appropriate software or firmware, the processor **701** is responsible for such tasks such as optimization. Various specially configured devices can also be used in place of a processor **701** or in addition to processor **701**. The complete implementation can also be done in custom hardware. The interface **711** is typically configured to send and receive data packets or data segments over a network. Particular examples of interfaces the device supports include Ethernet interfaces, frame relay interfaces, cable interfaces, DSL interfaces, token ring interfaces, and the like.

[0071] In addition, various very high-speed interfaces may be provided such as fast Ethernet interfaces, Gigabit Ethernet interfaces, ATM interfaces, HSSI interfaces, POS interfaces, FDDI interfaces and the like. Generally, these interfaces may include ports appropriate for communication with the appropriate media. In some cases, they may also include an independent processor and, in some instances, volatile RAM. The independent processors may control such communications intensive tasks as packet switching, media control and management.

[0072] According to particular example embodiments, the system **700** uses memory **703** to store data and program instructions and maintain a local side cache. The program instructions may control the operation of an operating system and/or one or more applications, for example. The memory or memories may also be configured to store received metadata and batch requested metadata.

[0073] Because such information and program instructions may be employed to implement the systems and/or methods described herein, the present invention relates to tangible, machine readable media that include program instructions, state information, etc. for performing various operations described herein. Examples of machine-readable media include hard disks, floppy disks, magnetic tape, optical media such as CD-ROM disks and DVDs; magneto-optical media such as optical disks, and hardware devices that are specially configured to store and perform program instructions, such as read-only memory devices (ROM) and programmable read-only memory devices (PROMs). Examples of program instructions include both machine code, such as produced by a compiler, and files containing higher level code that may be executed by the computer using an interpreter.

[0074] As described in various examples of the present disclosure, a supertorrefaction process can be used to convert raw biomass into biocarbon product. Because the raw biomass would otherwise decay and release harmful amounts of CO₂ into the environment, converting this raw biomass to biocarbon product can reduce net carbon emissions to the environment. The processes described in the present disclosure provide several benefits over traditional torrefaction, making it an attractive option for biomass conversion.

[0075] According to various embodiments, supertorrefaction can produce a range of biocarbon products, including biocoals, biochar, inert carbon and activated carbon. The range of products may include others that result from interactions of the salts with the biomass and operating conditions. In some example, the biocoal has a higher energy density compared to conventional torrefaction products. Depending on the operating parameters, biochar produced through supertorrefaction has increased reactivity, making it more suitable for various applications, including soil amendment and carbon sequestration. Furthermore, it can effectively capture and retain nutrients and water, enhancing its value in agriculture. Depending on the operating parameters the biocarbon product can also have very high level of inert carbon, suitable for carbon sequestration for up to 1000 years.

[0076] Additionally, supertorrefaction tends to produce biochar with lower tar and volatile organic compound (VOC) content compared to torrefaction. This is achieved by effectively removing these compounds during the process, resulting in cleaner biochar and reduced emissions. Reduced tars also prolong the equipment lifespan and save on labor to clean the equipment. Supertorrefaction can process a wider range of biomass feedstocks, including those with high moisture content, more effectively than traditional torrefaction. This broader feedstock flexibility can reduce waste and increase biomass utilization. In many respects supertorrefaction biocarbon products resemble those from fast pyrolysis more than those from conventional torrefaction.

[0077] While the present disclosure has been particularly shown and described with reference to specific embodiments thereof, it will be understood by those skilled in the art that changes in the form and details of the disclosed embodiments may be made without departing from the spirit or scope of the invention. Specifically, there are many alternative ways of implementing the processes, systems, and apparatuses described. It is therefore intended that the invention be interpreted to include all variations and equivalents that fall within the spirit and scope of the present invention. Moreover, although particular features have been described as part of each example, any combination of these features or additions of other features are intended to be included within the scope of this disclosure. Accordingly, the embodiments described herein are to be considered as illustrative and not restrictive.

[0078] Furthermore, in the foregoing specification, various techniques and mechanisms may have been described in singular form for clarity. However, it should be noted that some embodiments include multiple iterations of a technique or multiple instantiations of a mechanism unless otherwise noted. For example, a system uses a basket in a variety of contexts but can use multiple baskets while remaining within the scope of the present disclosure unless otherwise noted. Similarly, various techniques and mechanisms may have been described as including a connection between two entities. However, a connection does not necessarily mean a direct, unimpeded connection, as a variety of other entities (e.g., pipes, channels, sensors, control mechanisms, switches, gates, valves, etc.) may reside between the two entities.

[0079] Additionally, in the foregoing specification, reference was made in detail to specific embodiments including one or more of the best modes contemplated by the inventors. While various embodiments have been described in the present disclosure, it should be understood that they have been presented by way of example only, and not limitation. Particular embodiments may be implemented without some or all of the specific details described. In other instances, well known process operations have not been described in detail in order not to unnecessarily obscure the mechanisms and techniques disclosed. Accordingly, the breadth and scope of the present

application should not be limited by any of the embodiments described, but should be defined in accordance with the claims and their equivalents.

Claims

1. A system comprising a batch process cooker for thermal processing of biomass, the batch process cooker configured to hold the biomass for high-temperature conversion into biocarbon products; a biomass basket disposed within the batch process cooker, wherein the biomass basket is configured to hold the biomass during thermal processing; a plenum lid integrated into the batch process cooker, the plenum lid including a plurality of distribution channels for evenly distributing molten salts onto the biomass within the biomass basket; a molten salt reservoir positioned below the batch process cooker, the molten salt reservoir storing molten salts utilized for heating and converting the biomass within the biomass basket; and a molten salt pump configured to pump molten salts from the molten salt reservoir to the plenum lid.
2. The system of claim 1, wherein thermal processing of biomass comprises supertorrefaction at tunable temperatures and time periods with variety molten salt mixtures to produce a range of biocarbon products.
3. The system of claim 1, wherein the biomass basket includes a robotic arm attachment for loading and unloading the biomass basket to and from the batch process cooker.
4. The system of claim 1, wherein the molten salts correspond to a molten salt mixture comprised of any eutectic mixture of individual salts.
5. The system of claim 1, wherein the batch process cooker further comprises a temperature control mechanism for regulating the temperature within the cooker during thermal processing.
6. The system of claim 1, wherein the plenum lid includes an adjustable flow control mechanism for fine-tuning the distribution of molten salts onto the biomass.
7. The system of claim 1, wherein the molten salt reservoir features safety interlocks and emergency shutdown mechanisms for ensuring safe operation.
8. The system of claim 1, further comprising a bypass system between the molten salt reservoir and the plenum lid, the bypass system enabling the recirculation of pumped molten salts to the molten salt reservoir when the biomass is being loaded into or biocarbon product is being removed from the batch process cooker.
9. The system of claim 1, wherein the plenum lid includes heat insulation to protect against heat loss and maintain efficient thermal processing.
10. The system of claim 1, further comprising a sluice washer configured to remove salts from the processed biocarbon product.
11. The system of claim 1, wherein the batch process cooker is a cylindrical cooker, and the biomass basket is a cylindrical basket.
12. The system of claim 11, wherein the cylindrical basket is a cylindrical vessel with solid vertical walls and a mesh bottom that allows the biomass to remain in the cylindrical basket while molten salts pass through the mesh bottom.
13. A sluice washer apparatus for washing biocarbon product or similar materials, comprising: a trough-shaped structure configured to receive salts and other impurities resulting from washing the biocarbon product; a water inlet system configured to supply water to the trough-shaped structure from a first end of the trough-shaped structure; a drain system at a second end of the trough-shaped structure for collecting water, salts, and other impurities; a biocarbon product basket disposed above the trough-shaped structure, wherein water is also injected onto the biocarbon product basket; a drive mechanism for moving the biocarbon product basket along the length of the trough-shaped structure; and a control system for regulating water flow and movement of the biocarbon product basket.
14. The sluice washer of claim 13, wherein the biocarbon product basket is moved from the second

- end of the trough-shaped structure to the first end of the trough-shaped structure during washing of the biocarbon product.
- 15.** The sluice washer of claim 13, wherein the trough-shaped structure is inclined to facilitate the controlled flow of water, biocarbon product, and saline solution from the first end to the second end.
- 16.** The system of claim 13, wherein the molten salts correspond to a molten salt mixture comprised of any eutectic mixture of individual salts.
- 17.** The sluice washer of claim 13, wherein the water inlet system includes flow control mechanisms for adjusting the water flow rate into the trough-shaped structure and into the biocarbon product basket.
- 18.** The sluice washer of claim 13, wherein the sluice washer is integrated into a system having multiple lines, the multiple lines including multiple sluice washers and multiple condenser systems associated with a single batch process cooker.
- 19.** The sluice washer of claim 13, wherein the sluice washer includes drive mechanism for moving the biocarbon product basket comprises a conveyor system and a drive bar, allowing for smooth and controlled movement of the biocarbon product basket.
- 20.** The sluice washer of claim 13, wherein the biocarbon product basket is a cylindrical vessel with solid vertical walls and a mesh bottom that allows the biocarbon product to remain in the biocarbon product basket while the salts pass through the mesh bottom.
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